UNIVERSITY OF ESWATINI

Department of Computer Science

Practices in Software Engineering I

Submitted to Dr. E. L Dube, Lecturer

In partial fulfilment of the requirements for CSC392: Practices in Software Engineering I

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UNESWA Fitness Gym Management System Documentation

Introduction

The UNESWA Fitness Gym Management System is designed to modernize the operations of the UNESWA Fitness Gym, located at the University of Eswatini (UNESWA) Main Campus. Previously, the gym relied on a manual filing system to manage member information, categorize payments, and track usage, which proved time-consuming, error-prone, and difficult to scale. This documentation outlines the development of a digital solution to address these challenges, enhancing efficiency, accuracy, and user experience for members, staff, and administrators. The system aims to streamline member registration, payment tracking, facility bookings, and reporting, providing a centralized and secure platform to support the gym's growing membership.

Specification

This section details the requirements and specifications for the UNESWA Fitness Gym Management System, ensuring it meets the needs of all stakeholders.

User Roles

Administrator: Oversees system configuration, manages user accounts, generates reports, and ensures data integrity.

Trainer: Schedules and conducts fitness sessions, monitors member progress, and ensures safe equipment usage.

Member: Registers for membership, books sessions, views payment status, and updates personal information.

Data Fields

Personal Information: Full name, surname, date of birth, gender, PIN, address, contact numbers, email, occupation, employer.

Membership Details: Category (e.g., UNESWA Student, Staff, Senior Citizen), status (Active, Pending Renewal, Inactive), payment method (Cash, Stop Order).

Activity Preferences: Swimming, weights, aerobics, cardio.

Payment History: Amount, date, method, receipt verification.

2.3 Functional Requirements

Member Management: Enable registration, categorization, and profile updates.

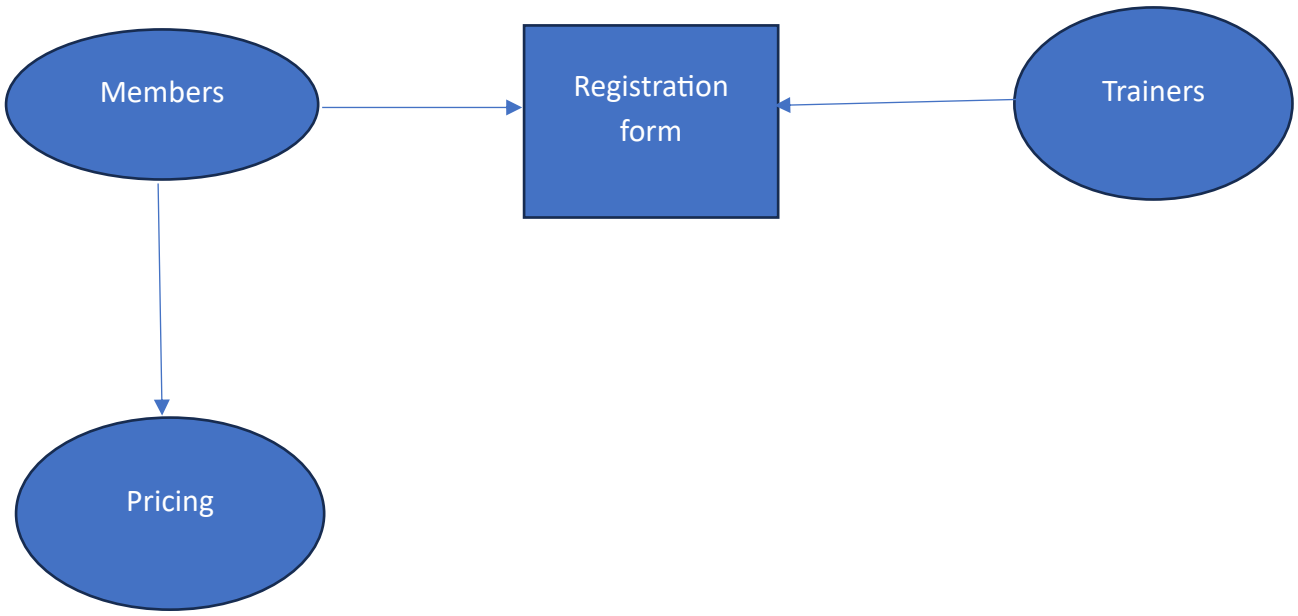
Payment Processing: Track and manage payments, generate receipts, and flag overdue accounts.

Booking System: Allow members to reserve sessions and facilities (e.g., gym, pool) in advance.

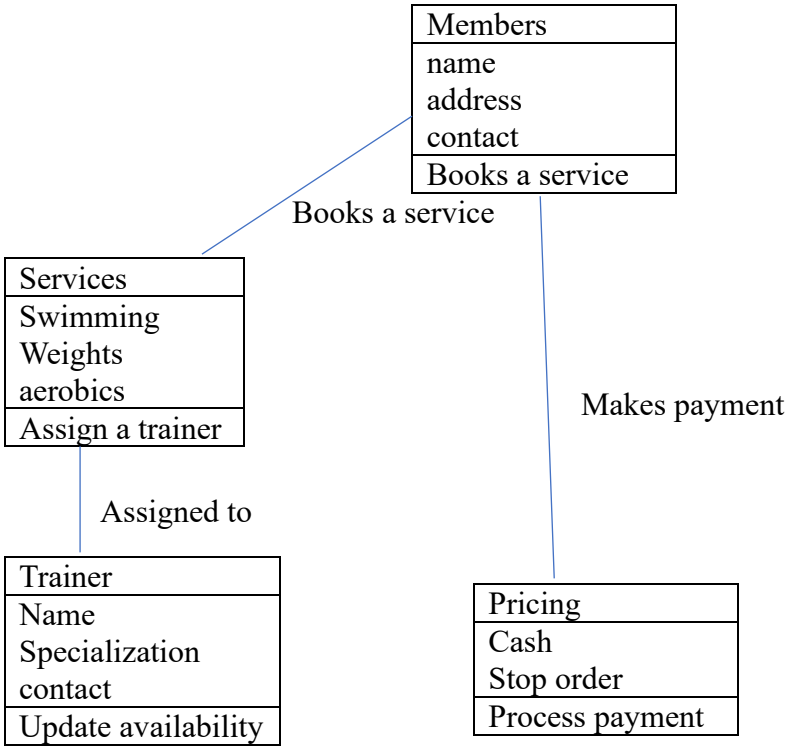
Reporting: Provide administrators with insights on membership trends, payment statuses, and usage statistics.

Security: Ensure data is stored securely with role-based access control.

Data Flow Diagram



UML Diagram



Design

This section outlines the system's architecture and implementation details.

Architecture

The system is a web-based application with the following components:

Frontend: Built using HTML for structure, CSS for styling, and JavaScript for interactivity.

Backend: Implemented with Node.js and Express to handle server-side logic and API endpoints. -

Database: MySQL is used to store member data, bookings, and payment records securely.

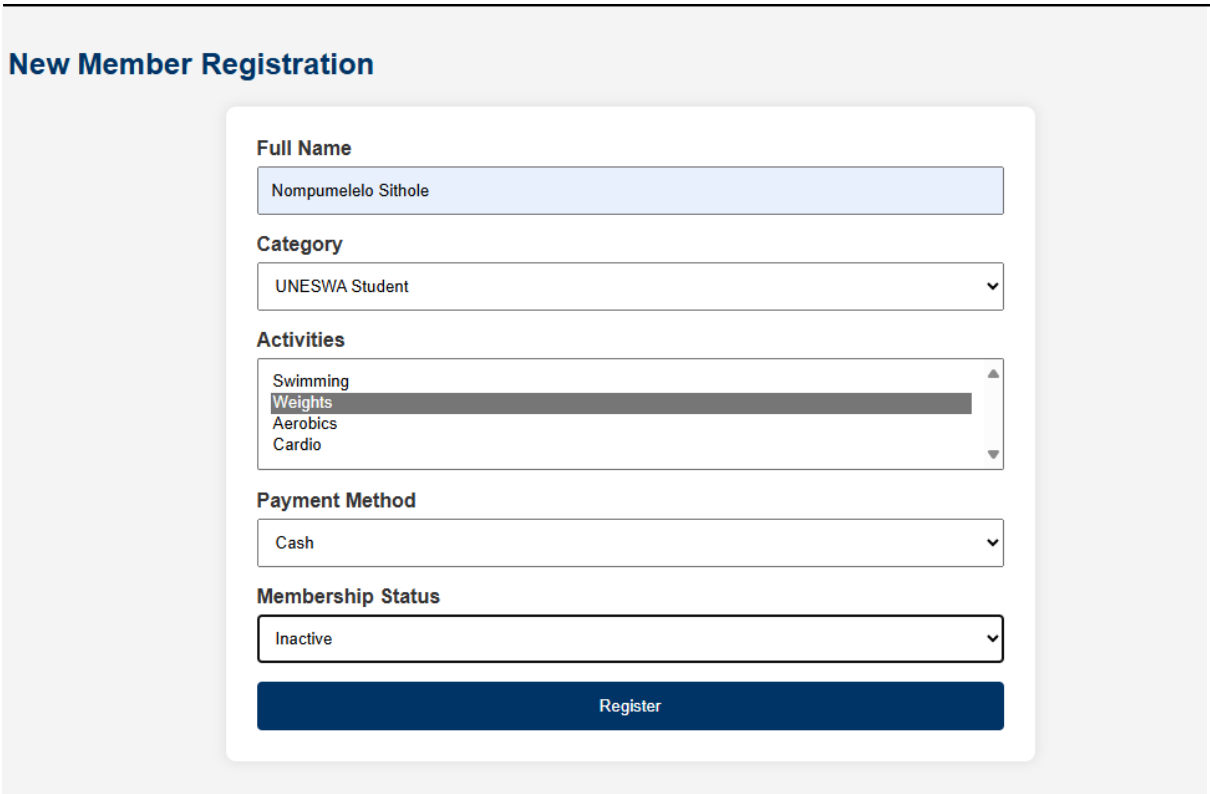
User Interface

The interface mirrors the structure of the provided HTML files (Home, About, Services, Pricing, Members, Trainers, Contacts), with dynamic forms for registration, booking, and payment management.

Testing

Unit Tests

Member Registration: Validate input fields (e.g., name, contact) and check for duplicate entries.



This screenshot displays the New Member Registration form used to register individuals at the gym facility.

The form includes fields for:

- Full Name
- Category (e.g., UNESWA Student)
- Activities (multi-select dropdown)
- Payment Method
- Membership Status

Test Inputs:

- **Full Name:** Nompumelelo Sithole
- **Category:** UNESWA Student
- **Activities Selected:** Swimming, Weights
- **Payment Method:** Cash
- **Membership Status:** Inactive

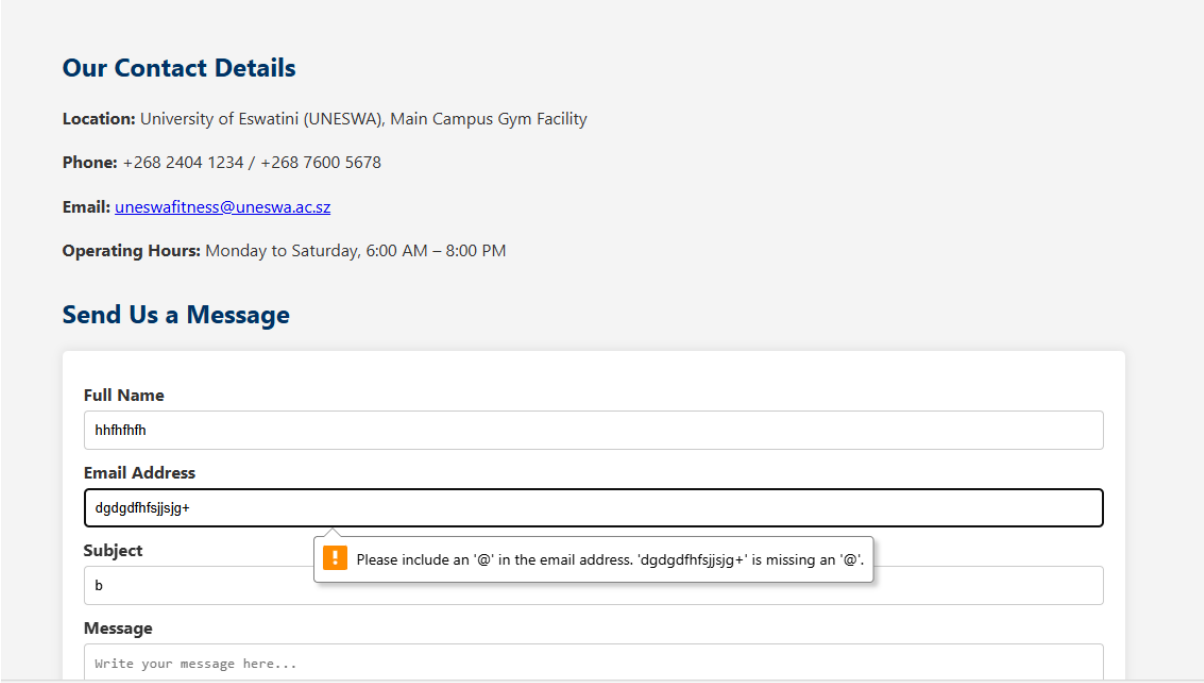
Expected Result: The form should allow entry of all details, support multi-selection for activities, and permit registration when required fields are filled.

Observed Result: All fields are correctly filled, and the "Register" button is enabled.

Test Outcome: Passed – The form accepts valid input, correctly handles multi-select options, and is ready for submission.

Payment Processing: Verify amount calculations and receipt generation.

Booking System: Test session availability and conflict detection.



This screenshot shows the contact form on the University of Eswatini Main Campus Gym Facility website during testing. It highlights email validation functionality. The tester entered an invalid email address (dgdgdfhfsjjsjg+) that lacks the required @ symbol.

Expected Result:
The system should detect the invalid email format and provide a clear error message.

Observed Result:
An error tooltip appears below the email field stating:

"Please include an '@' in the email address. 'dgdgdfhfsjjsjg+' is missing an '@'."

Integration Tests

End-to-End Registration: Confirm member data flows from form to database.

Booking and Payment Linkage: Ensure bookings are tied to valid payment records.

Report Generation: Validate accuracy of membership and payment reports.

Performance Tests

Simulate 500 concurrent users to assess system responsiveness.

Measure load time for report generation with 1,000 records.